

DO NOT WRITE ON TEST



**FINANCIAL MATH &
ANALYSIS CONCEPTS**

~OPEN EVENT~

(190)

NATIONAL 2023

TOTAL POINTS

_____ **100 Points**

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.
4. For any interest calculations, assume a 360-day year.
5. Unless otherwise stated, assume that all plant assets were purchased on January 1.
6. Unless otherwise stated, assume that the overtime rate is 1.5 regular time pay for any hours worked in excess of 40 hours per week.

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

1. Van Durkes is an accountant who works for a temporary help agency during tax season. He is paid \$259 per day. Last month he worked 19 days as a temporary accountant. What was his gross pay for the month?
 - A. \$3,108
 - B. \$4,921
 - C. \$5,439
 - D. \$5,698
2. On April 30, Fern's bank statement balance was \$516.24. Her check register showed that a deposit of \$382.10 was outstanding. The following checks were also outstanding: #108, \$45.93; #109, \$108.12; #111, \$6.87. What was the corrected bank balance?
 - A. \$134.14
 - B. \$295.06
 - C. \$525.71
 - D. \$737.42
3. Heng-Che Pai's credit card statement for last month showed a previous balance of \$289.16, new purchases of \$107.99, a membership fee of \$35, a finance charge of \$5.96, and a payment of \$100. What is the new balance on this credit card?
 - A. \$189.16
 - B. \$253.21
 - C. \$338.11
 - D. \$438.11
4. A company bought a site license for a software that retails for \$49.99 per copy. They paid \$134,486 for use on their 3,200 computer workstations. How much did they save by buying the site license?
 - A. \$25,482
 - B. \$18,346
 - C. \$9,234
 - D. \$2,548
5. The Gutierrez family want to buy a home. They estimate these home operating expenses: property tax, \$3,400; insurance, \$485; utilities, \$1,480; maintenance, \$1,200; mortgage interest, \$6,400; lost interest on down payment, \$420. Their estimated income tax savings are \$1,800. What will be the net cost of owning the home in the first year?
 - A. \$9,575
 - B. \$10,786
 - C. \$11,585
 - D. \$13,385

6. Aubrey had \$140 in change in her cash register at the start of work. At the end of her work period, there was \$2,961.19 in the cash register. The register totals showed that she received \$2,983.27 and paid out \$159.76 during the period. How much was the cash over or short?
- A. \$2.32 over
 - B. \$2.32 short
 - C. \$137.68 over
 - D. \$137.68 short
7. Daniel earned \$6,200 last month. He paid an overall FICA tax rate of 7.65% that consists of a 6.2% social security tax and a 1.45% Medicare tax. Federal income taxes withheld from his wagers were \$1,512. What total amount of taxes was deducted from his pay last month?
- A. \$474.30
 - B. \$1,240.25
 - C. \$1,986.30
 - D. \$4,213.70
8. The cost of an advertising insert in a daily newspaper delivered to one zip code is \$1,600 per edition. There are 7,920 households who receive the newspaper. What will be the cost of the ad per household?
- A. \$.20
 - B. \$.24
 - C. \$.32
 - D. \$.45
9. A reduction of the invoice price given for early payment is known as a _____.
- A. list price
 - B. sales invoice
 - C. trade discount
 - D. cash discount
10. An invoice for \$3,410 dated May 1 has terms of sale of 2/10, 1/20, n/60. What is the amount of the cash price if the invoice is paid on May 14?
- A. \$3,410
 - B. \$3,380
 - C. \$3,375.90
 - D. \$3,341.80
11. A discount series includes a trade discount that has two or more discounts.
- A. True
 - B. False

12. Two programmers developed a 6,800-line program. One programmer spent 380 hours on the program and was paid \$21.50 per hour. The other spent 370 hours and was paid \$17.50 an hour. What was the average programming cost per line for the program?
- A. \$1.45
 - B. \$2.15
 - C. \$3.59
 - D. \$5.25
13. Maria bought 500 shares of preferred stock at \$12.87. The commission on the purchase was \$184.38. Maria later sold stock at \$20.15. The commission on the sale was \$228. What was the amount of profit or loss on the sale?
- A. \$1,245.22 loss
 - B. \$1,245.22 profit
 - C. \$3,227.62 loss
 - D. \$3,227.62 profit
14. Ezra wants to earn 15% annual net income on his \$10,000 cash investment in property. Her annual expenses for owning the property are \$2,700. What monthly rent must Ezra charge to make her earning goal?
- A. \$350
 - B. \$450
 - C. \$550
 - D. \$650
15. Jasmine plans to rent an apartment for \$710 a month, including heat and water. The security deposit will be one-month's rent. She estimates other annual costs connected with the apartment will be insurance \$110; utilities, \$600; covered parking space, \$240. What is Jasmine's first-year cost of renting the apartment?
- A. \$8,520
 - B. \$9,340
 - C. \$10,180
 - D. \$12,250
16. Joshua had a new 36 square yard wall-to-wall rug installed in his home. The rug cost \$15.99 a square yard, the padding cost \$3 a square yard and the rug was installed for \$4 a square yard. The sales tax rate is 4.9% and is not applied to the labor. What was the total cost of the purchase?
- A. \$144
 - B. \$575.64
 - C. \$683.64
 - D. \$861.14

17. An employer provides dental health insurance to employees. The monthly premium cost per employee is \$36. If the employees pay 65% of the premium, what is the total annual premium paid by an employee for the dental insurance?
- A. \$140.40
 - B. \$280.80
 - C. \$432.00
 - D. \$712.80
18. Upon graduation, Pedro redeemed 300 shares of a mutual fund at \$17.22. His total investment in the shares was \$4,391.10. What was his profit or loss?
- A. \$774.90 profit
 - B. \$774.90 loss
 - C. \$924.90 profit
 - D. \$924.90 loss
19. Flavian's check register showed a balance of \$782.29 at the beginning of the day. During the day, he used his debit card three times in the amounts of \$89.39, \$585.87, and \$193.92. He also transferred \$200 from his savings to put in his checking account. What was the balance of Flavian's check register at the end of the day?
- A. (\$313.11)
 - B. \$313.11
 - C. (\$113.11)
 - D. \$113.11
20. The point at which a business will make no profit or suffer no loss is the _____, which is the point that income from sales equals the total cost of producing and selling goods.
- A. surplus
 - B. deficit
 - C. break-even point
 - D. equilibrium
21. A van costing \$16,000 is estimated to depreciate 20% each year. What is the estimated book value at the end of the first year?
- A. \$3,200
 - B. \$12,800
 - C. \$14,200
 - D. \$15,680
22. The Birgams' gross income for a year was \$79,104. From that amount, they subtracted a standard deduction of \$9,700, and 3 exemptions at \$3,100 each to find their taxable income. What was their taxable income total?
- A. \$60,104
 - B. \$64,456
 - C. \$66,304
 - D. \$69,404

23. Rossco Products sold these items to a local flooring company: 60 cases of tile @ \$24.54 a case, 12 gallons adhesive @ \$31.89 a gallon, and 12 tile cutters @ \$48.23 each. What was the total of the sales invoice before tax?
- A. \$578.76
 - B. \$961.44
 - C. \$1,472.40
 - D. \$2,433.84
24. A retailer bought shoes to sell for \$120 and sold those shoes for \$162. If the markup is based on cost, what markup percentage is the retailer charging?
- A. 25%
 - B. 30%
 - C. 35%
 - D. 40%
25. Lynn is paid \$2.08 for every usable machine part she makes. During one week, she made 220 parts, 14 of which were unusable. What was Lynn's gross pay for the week?
- A. \$357.60
 - B. \$428.48
 - C. \$457.60
 - D. \$504.56
26. Stuart listed these items on his checking account deposit slip: (bills) 47 ones, 3 fives, 2 tens, and 6 twenties; (coins) 129 quarters, 74 dimes, 22 nickels; one check for \$45.50. He received cash of one one-hundred dollar bill. What was the total of Stuart's deposit?
- A. \$188.25
 - B. \$242.75
 - C. \$288.25
 - D. \$301.75
27. You have \$500 on deposit in a one-year certificate of deposit that pays 6% compounded quarterly. What is the effective annual interest rate, to the nearest tenth percent?
- A. 5.8%
 - B. 6.1%
 - C. 6.4%
 - D. 7.1%
28. What is the gross earnings for an employee who works 43 hours and is paid \$11.23/hour?
- A. \$449.20
 - B. \$482.89
 - C. \$499.74
 - D. \$521.67

29. Judd makes a regular time pay rate of \$12.60. He gets time-and-a-half for overtime and double time for work on Saturdays and Sundays. He worked the following schedule last week: Monday 8 hours, Tuesday 10 hours, Wednesday 10 hours, Thursday 8 hours, Friday 10 hours, Saturday 4 hours. What was his gross pay for the week?
- A. \$504.00
 - B. \$613.40
 - C. \$718.20
 - D. \$812.50
30. Find the property tax due on a building assessed at \$360,000 if the tax rate is \$37.40 per \$1,000.
- A. \$12,456.25
 - B. \$12,599.00
 - C. \$13,256.25
 - D. \$13,464.00
31. Brody bought a new car for \$28,000. He plans to drive the car for a total of 100,000 miles over the course of the next six years. At the end of six years, Brody estimates he can sell the car for \$4,200. If Brody drove the car 18,600 miles in the first year, how much depreciation will he record that year using the units-of-production method?
- A. \$5,208.00
 - B. \$4,426.80
 - C. \$4,666.67
 - D. \$3,966.67
32. Yang Le bought 500 shares of stock at a total cost of \$12,762. She sold the shares at \$40.76 and was charged a commission and other fees of \$374. What was her profit or loss from the sale of the stock?
- A. \$7,618 profit
 - B. \$7,618 loss
 - C. \$7,244 profit
 - D. \$7,244 loss
33. Laura's pension fund pays her 1.9% of her average salary for the last three years for each year of service. In the last three years, she earned \$57,000, \$59,800 and \$62,500. She will be retiring after 30 years of service. What monthly pension will she receive?
- A. \$2,838.92
 - B. \$2,945.54
 - C. \$3,260.24
 - D. \$4,201.67

34. Lateesha had an adjusted gross income of \$36,057. She is single and her itemized deductions total \$3,950. She claimed 1 exemption for herself at \$3,100. What is her taxable income?
- A. \$22,506
 - B. \$29,007
 - C. \$36,057
 - D. \$43,107
35. HighTech Stereo, Inc. has 1,308 computer users. If they provide a ratio of technical support personnel of 1 for every 60 users, how many technical support personnel do they have, to the nearest whole person?
- A. 20
 - B. 21
 - C. 22
 - D. 24
36. The Stalcup Specialty Store purchased a new cash register for \$5,000. Stalcup expects to use the cash register for eight years and expects to sell it for \$800 at the end of that time. Using straight-line depreciation, how much will Stalcup depreciate the cash register each year?
- A. \$725
 - B. \$675
 - C. \$625
 - D. \$525
37. The Keller Company runs a newspaper employment ad each Sunday. The ad should cost \$1,400 a day. Because of the number of ads it runs, Keller Company receives a 22% discount off the usual price. What is the cost for Keller Company to run ads for one year?
- A. \$1,092
 - B. \$13,104
 - C. \$26,240
 - D. \$56,784
38. Debt-to-equity ratio is found by dividing the long-term liabilities by the firm's capital.
- A. True
 - B. False

39. Davis & Co., a distributor, purchased generic t-shirts from different manufacturers during the month of March.

March 1 700 @ \$7.30
 March 14 1,200 @ \$6.90
 March 18 300 @ \$7.62
 March 28 500 @ \$7.41

The unit value of the beginning inventory of 600 t-shirts was \$7.28. Find the value of the ending inventory of 700 t-shirts using FIFO.

- A. \$5110
 - B. \$5,229
 - C. \$6,125
 - D. \$7,150
40. A gallon of low-fat milk is priced at \$2.98. A half-gallon of the same milk is \$1.89. How much can be saved by buying 1 gallon of milk instead of buying two half-gallons?
- A. \$.70
 - B. \$.80
 - C. \$1.09
 - D. \$1.19
41. You signed a promissory note at your credit union to borrow \$4,200 for 6 months at 16% annual interest. What was the total amount of interest owed on the note?
- A. \$145
 - B. \$244
 - C. \$336
 - D. \$672
42. Recently, Argentina had exports of \$29.57 billion and imports of \$13.27 billion. How much of a deficit or surplus did they have?
- A. \$13.27 billion deficit
 - B. \$13.27 billion surplus
 - C. \$16.3 billion deficit
 - D. \$16.3 billion surplus
43. The Rodriguez family saves \$.25 per liter by buying orange juice in bulk. The family uses 2.5 L of orange juice a week. How much will the Rodriguez family save in a year by buying juice in bulk?
- A. \$32.50
 - B. \$52.40
 - C. \$61.26
 - D. \$72.32

44. Felicia spent $2\frac{1}{2}$ days at a business conference. She was reimbursed for traveling the 173 miles to and from the show by personal car at \$.36 a mile. She was also given a \$146 per diem per full day (Felicia does not receive a per diem for partial days). What was her total reimbursement for the conference?
- A. \$365.00
 - B. \$416.56
 - C. \$472.25
 - D. \$480.60
45. An equipment company uses the Web to advertise on the Internet. Their web-hosting firm charged them \$3,150 to develop their web site and \$68.50 a month to maintain it. They also charge \$20 annually to register the company's web site address and \$65 a month for web server space and an Internet connection. What is the total cost for the first year to the company?
- A. \$3,303
 - B. \$3,772
 - C. \$4472
 - D. \$4772
46. LaVina insured her home for \$116,000 at an annual rate of \$.42 per \$100. What is her annual premium to the nearest dollar?
- A. \$420
 - B. \$467
 - C. \$487
 - D. \$520
47. The purchase of a 3-year old used car is \$12,450. Other costs include registration fees of \$128 and sales tax of 4%. The buyer made a down payment of \$3,800. What is the balance due of the car?
- A. \$9,276
 - B. \$10,568
 - C. \$11,250
 - D. \$13,076
48. Ella invested \$5,000 in a 4-year CD that paid 5.2% annual interest. When she withdrew \$800 at the end of 3 years, her early withdrawal penalty was 6 months' interest. What was the amount of the penalty?
- A. \$20.80
 - B. \$60.20
 - C. \$80.00
 - D. \$779.20

49. Terry works on a piece-rate basis. He completed 70 pieces on Monday, 68 on Tuesday, 74 on Wednesday and 72 on Thursday. He is paid \$1.20 for each piece. How many pieces must he complete on Friday so that his earnings for the 5 days will average \$84 a day?
- A. 60
 - B. 66
 - C. 70
 - D. 76
50. BoldLine Inc. plans to make patio chairs and sell them at \$75 each. They estimate their fixed costs to produce the chairs at \$300,000 and variable costs of \$45 per chair. How many chairs must BoldLine Inc. sell to break even?
- A. 1,900
 - B. 4,500
 - C. 7,900
 - D. 10,000